

Unit 1

ATOMIC STRUCTURE

62 items · 182 marks

Syllabus sections in this unit

- 1.1 Particles in the atom and atomic radius
- 1.2 Isotopes
- 1.3 Electrons, energy levels and atomic orbitals
- 1.4 Ionisation energy

Unit 1: Atomic structure**1. M/J 26/P21/Q1(b)**

(b) Table 1.1 shows information about sulfur and chlorine.

Table 1.1

element	electronic configuration	first ionisation energy /kJ mol ⁻¹
sulfur	[Ne] 3s ² 3p ⁴	1000
chlorine	[Ne] 3s ² 3p ⁵	1260

(i) Name the orbital that contains the lowest energy electron in an atom of sulfur. Draw the shape of this orbital.

name

shape

[1]

(ii) Construct an equation to represent the first ionisation energy of chlorine.

[1]

(iii) Explain the difference in the values of first ionisation energy of sulfur and chlorine in Table 1.1.

[3]

2. M/J 26/P22/Q1(a)

Context

1 Oxygen reacts with many elements.

(a) (i) State the total number of electrons that occupy the 2p sub-shell in an oxygen atom.

[1]

(ii) Name the orbital that contains the highest energy electron in an oxygen atom. Draw the shape of this orbital.

name

shape

[1]

3. M/J 26/P22/Q2(a)

- 2 (a)** Protons, neutrons and electrons are released with the same velocity from a source.

Figure 2.1 shows the movement of two of these types of particle when they travel between oppositely charged plates.

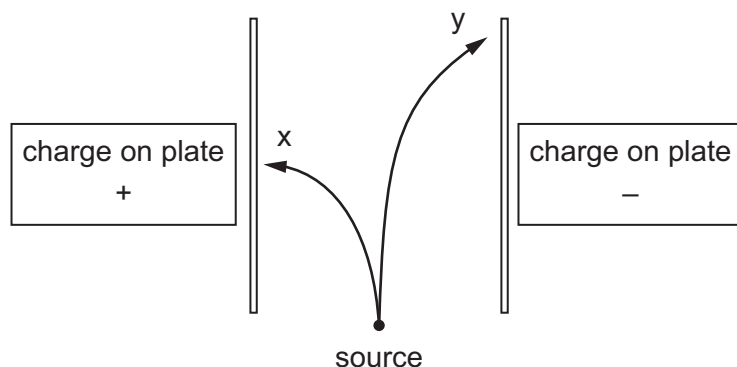


Figure 2.1

- (i)** Name each type of particle that produces the deflection shown by curves x and y.

curve x =

curve y =

[1]

- (ii)** Suggest why the particles producing curve y are deflected less than the particles producing curve x.

[1]

4. M/J 26/P22/Q2(b)

- (b)** A student considers the information in Table 2.1 and proposes hypothesis 1.

hypothesis 1 'Atoms with a higher nuclear charge are always larger.'

Table 2.1

atom	atomic number	mass number	atomic radius/nm
A	6	12	0.077
B	17	37	0.099
C	14	28	0.117

- (i)** Identify a pair of atoms from Table 2.1 that supports hypothesis 1.

[1]

- (ii)** Identify a pair of atoms from Table 2.1 that shows hypothesis 1 is false.

[1]

- (iii)** Explain why atom **A** is smaller than atom **C**.

[2]

5. M/J 26/P23/Q2(a)

Context

- 2** Group 18 elements exist as colourless gases at room temperature and pressure.

Table 2.1

atom	relative volume of atom
neon	172
argon	287
krypton	355
xenon	446

- (a) Use Table 2.1 to describe and explain the variation in atomic radius down Group 18.

[2]

6. M/J 26/P23/Q5(b)(i)

Context

- 5** Propane, $\text{CH}_3\text{CH}_2\text{CH}_3$, reacts with $\text{Cl}_2(\text{g})$ in the presence of ultraviolet light to produce different organic products, including those shown in Table 5.1.

Table 5.1

structure of product	systematic name of product
$\text{CH}_3(\text{CH}_2)_2\text{Cl}$	1-chloropropane
$\text{CH}_3\text{CHClCH}_3$	2-chloropropane
$(\text{CH}_3)_2\text{CHCH}(\text{CH}_3)_2$	2,3-dimethylbutane
	hexane
	2-methylpentane

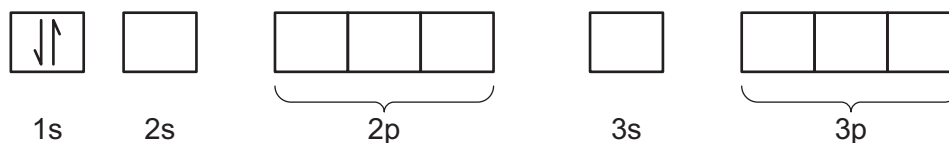
Context: part (b) (introduction)

- (b) The mechanism for this reaction involves three stages.

stage 1 initiation**stage 2** propagation**stage 3** termination

During the initiation reaction, a chlorine free radical is produced.

- (i) Complete Figure 5.1 to describe the arrangement of electrons in a chlorine free radical.

**Figure 5.1**

[1]

7. M/J 26/P24/Q1(a)

Context

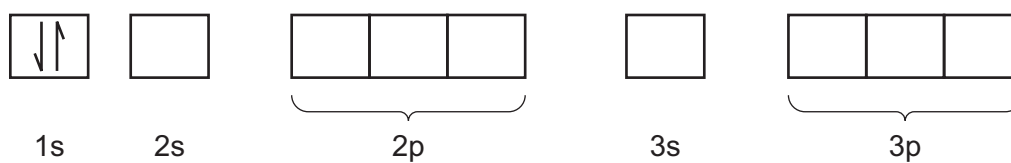
- 1** Aluminium is a Group 13 element that forms a number of compounds.

Table 1.1

atom or ion	number of protons	number of neutrons
^{27}Al		
$^{28}\text{Al}^{3+}$		

- (a) (i) Complete Table 1.1 to show the number of particles in each atom or ion. [2]

- (ii) Complete Figure 1.1 to show the electronic configuration of an aluminium atom.

**Figure 1.1**

[1]

8. MAR 26/P22/Q1(a)

Context

1 The Group 2 elements Mg to Ba show trends in their physical properties down the group.

(a) Table 1.1 shows some information about the elements Mg to Ba.

Table 1.1

element	proton number	shorthand electronic configuration	sum of first and second ionisation energies / kJ mol^{-1}	third ionisation energy / kJ mol^{-1}
Mg	12	[Ne] $3s^2$	2186	7740
Ca	20	[Ar] $4s^2$	1740	4940
Sr	38	[Kr] $5s^2$	1608	4120
Ba	56	[Xe] $6s^2$	1468	3390

(i) Give the full electronic configuration of a calcium atom.

[1]

(ii) Explain the trend shown in the sum of the first and second ionisation energies, as proton number increases.

[2]

(iii) Write an equation to represent the third ionisation energy of barium.

[1]

(iv) Explain why the third ionisation energy of magnesium is much greater than the sum of its first and second ionisation energies.

[2]

9. M/J 25/P21/Q2(a)

Context

2 A sample of iron is analysed using a mass spectrometer. The mass spectrum shows three isotopes of iron are present in the sample.

(a) Define isotopes.

[1]

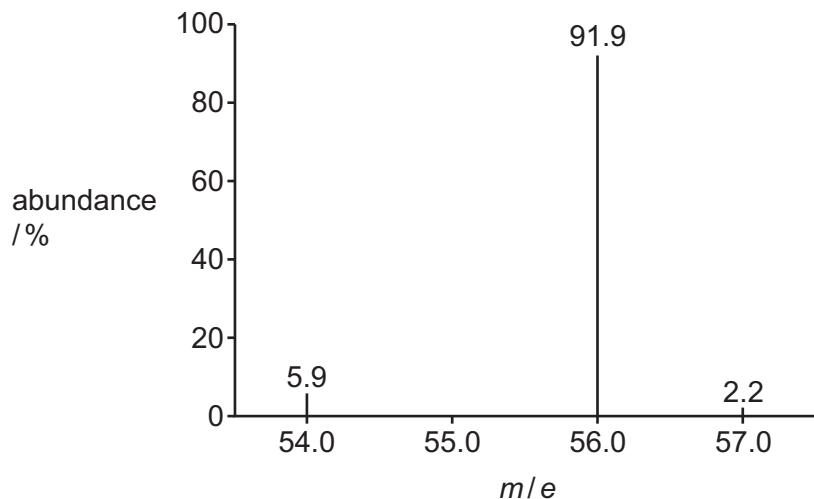
10. M/J 25/P21/Q2(b)(ii)

Context

- 2** A sample of iron is analysed using a mass spectrometer. The mass spectrum shows three isotopes of iron are present in the sample.

Context: part (b) (introduction)

- (b)** Fig. 2.1 shows the mass spectrum of the sample of iron.

**Fig. 2.1**

- (ii)** Complete Table 2.1 to show the number of protons and nucleons in one atom of ^{56}Fe .

Table 2.1

particle	number of particles in one atom of ^{56}Fe
protons	
nucleons	

[2]

11. M/J 25/P21/Q2(c)

Context

- 2** A sample of iron is analysed using a mass spectrometer. The mass spectrum shows three isotopes of iron are present in the sample.

- (c)** Deduce the number of pairs of electrons in the shell with principal quantum number $n = 3$ in an Fe atom.

number of pairs of electrons

[1]

12. M/J 25/P21/Q2(d)

Context

- 2** A sample of iron is analysed using a mass spectrometer. The mass spectrum shows three isotopes of iron are present in the sample.

- (d)** Write an equation to represent the first ionisation energy of iron.

[1]

13. M/J 25/P21/Q2(e)

Context

- 2** A sample of iron is analysed using a mass spectrometer. The mass spectrum shows three isotopes of iron are present in the sample.

(e) Suggest how the value for the first ionisation energy of ^{54}Fe compares to the first ionisation energy of ^{56}Fe . Explain your answer in terms of the factors that affect ionisation energy.

[4]

14. M/J 25/P22/Q2(b)

Context

- 2** A sample of iron contains three different isotopes and has a relative atomic mass, A_r , of 55.8.

(b) Deduce the number of pairs of electrons in the 3d sub-shell in an iron(II) ion.

[1]

15. M/J 25/P22/Q2(c)

Context

- 2** A sample of iron contains three different isotopes and has a relative atomic mass, A_r , of 55.8.

(c) Sketch the shape of the lowest energy orbital in the shell with principal quantum number $n = 2$.

[1]

16. M/J 25/P22/Q2(d)

Context

- 2** A sample of iron contains three different isotopes and has a relative atomic mass, A_r , of 55.8.

(d) Complete Table 2.2 to show information about particles in one ion of $^{56}\text{Fe}^{3+}$.

Table 2.2

particle	number of particles present in one ion of $^{56}\text{Fe}^{3+}$
electrons	
	30

[2]

17. M/J 25/P22/Q2(e)

Context

- 2** A sample of iron contains three different isotopes and has a relative atomic mass, A_r , of 55.8.

(e) The atomic radius of iron is $1.26 \times 10^{-10}\text{ m}$.

Suggest the change to the radius, if any, after an iron atom reacts to produce an Fe^{3+} ion. Explain your answer.

[2]

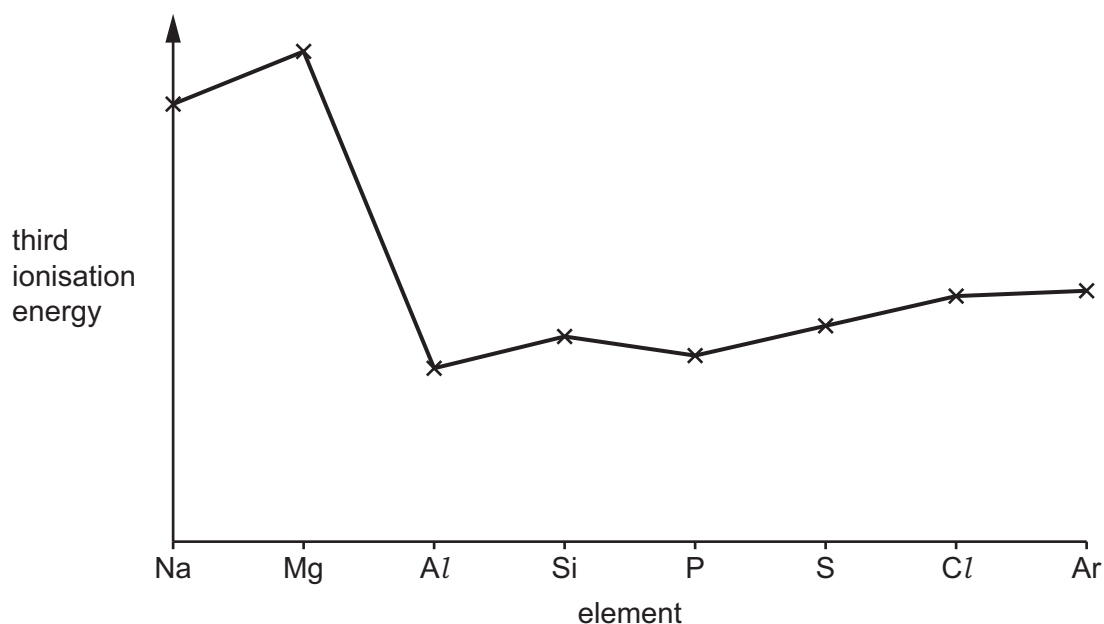


Fig. 1.2

- (i) Write an equation, including state symbols, to represent the **third** ionisation energy of argon. [2]
- (ii) The differences in the values for third ionisation energy shown in Fig. 1.2 are due to differences in the strength of attraction between the nucleus and the outer electron of each ion. State **two** factors that affect the strength of attraction between the nucleus and the outer electron. [2]
- 1
 - 2
- (iii) Use Fig. 1.2 to suggest the most significant factor that determines the size of attraction between the nucleus and the outer electron. Explain your answer. [2]
- (iv) In the third ionisation energy of argon, the ion produced has the electronic configuration $1s^2 2s^2 2p^6 3s^2 3p^3$. [2]

Complete Fig. 1.3 to show the arrangement of electrons in the orbitals of this ion.

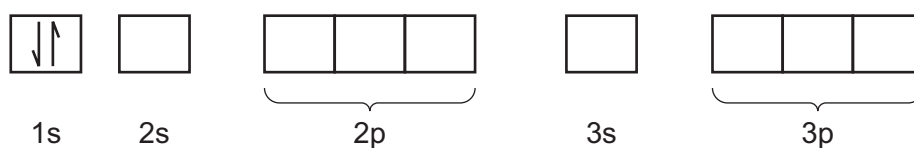


Fig. 1.3

[1]

21. MAR 25/P22/Q2(b)

(b) The Group 2 elements form stable 2+ cations.

(i) State and explain the variation in ionic radius of the Group 2 elements down the group.

[2]

(ii) Table 2.1 shows successive ionisation energy values for beryllium, Be.

Table 2.1

	1st	2nd	3rd	4th
ionisation energy / kJ mol^{-1}	900	1760	14 800	21 000

Use Table 2.1 to state and explain:

- the general trend in these values
- the significance of the large difference between the 2nd and 3rd ionisation energies.

[2]

22. MAR 25/P22/Q2(c)

(c) All the Group 2 elements except beryllium have more than one stable isotope.

(i) Beryllium exists as the single isotope ${}^9_4\text{Be}$.

Describe the distribution of mass within an atom of ${}^9_4\text{Be}$.

[2]

(ii) Complete Table 2.2 to show the numbers of protons and neutrons in the isotopes of magnesium.

Table 2.2

isotope	number of protons	number of neutrons
magnesium-24		
magnesium-25		
magnesium-26		

[2]

(iii) Fig. 2.1 shows the behaviour of a beam of protons in an electric field.

Complete Fig. 2.1 to show the behaviour of separate beams of neutrons and electrons in the same electric field.

Label your diagram clearly. Assume that the beams of each particle are moving at the same velocity.

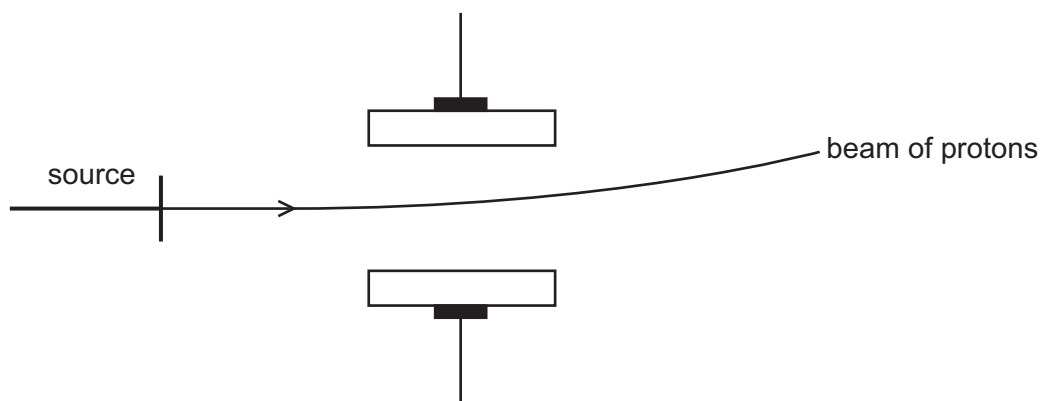


Fig. 2.1

[3]

23. O/N 24/P21/Q1(a)

Context

1 Cobalt, rhodium and iridium are metals in the same group of the Periodic Table.

(a) The shorthand electronic configuration of cobalt is $[\text{Ar}]3d^74s^2$.

(i) Identify what is meant by $[\text{Ar}]$ by giving its full electronic configuration.

[1]

(ii) The lowest-energy electrons in cobalt are in the 1s orbital.

Draw the shape of a 1s orbital.

[1]

(iii) Deduce the number of unpaired electrons in a cobalt atom.

[1]

24. O/N 24/P21/Q1(b)

Context

1 Cobalt, rhodium and iridium are metals in the same group of the Periodic Table.

(b) Table 1.1 gives some details of the stable naturally occurring isotopes of rhodium and iridium.

Table 1.1

isotope	number of protons	number of neutrons	total number of electron shells
$^{103}_{45}\text{Rh}$		58	
$^{191}_{77}\text{Ir}$			6
$^{193}_{77}\text{Ir}$			6

Complete Table 1.1.

[3]

25. O/N 24/P21/Q3(c)

Context

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

(c) Fig. 3.1 shows a sketch of some of the ionisation energies of silicon, Si.

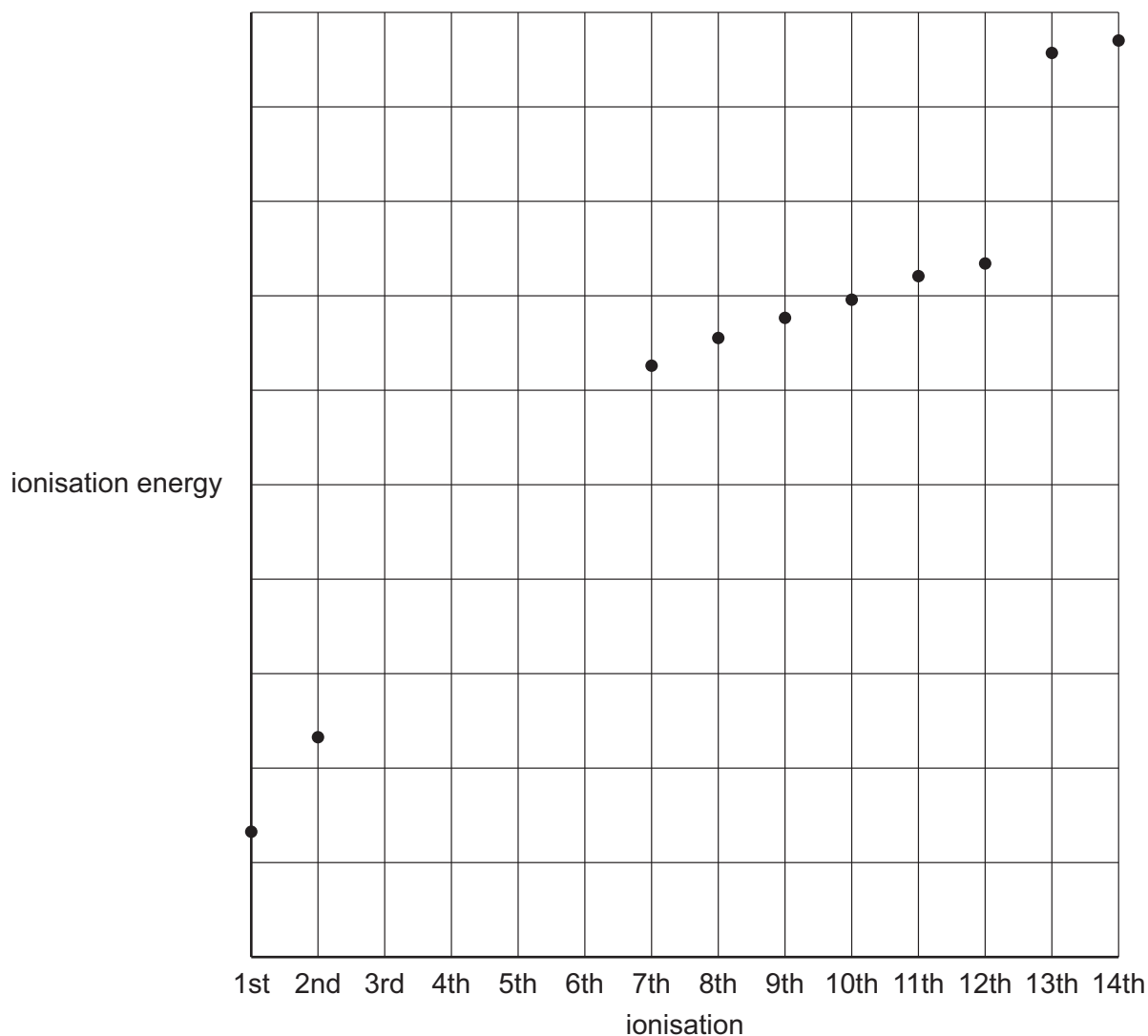


Fig. 3.1

(i) Complete the graph in Fig. 3.1 to show the third to sixth ionisation energies of Si.

[2]

(ii) Construct an equation to represent the second ionisation energy of Si.

[1]

26. O/N 24/P22/Q1(a)

Context

1 Vanadium, niobium and tantalum are metals in the same group of the Periodic Table.

(a) The shorthand electronic configuration of vanadium in the ground state is $[\text{Ar}]3d^34s^2$.

(i) State what is meant by the term ground state.

[1]

(ii) Show the electronic configuration of vanadium using electrons in boxes notation.



[1]

(iii) Deduce the total number of electrons in the p sub-shells of a vanadium atom.

[1]

27. O/N 24/P22/Q1(b)(i)

Context

1 Vanadium, niobium and tantalum are metals in the same group of the Periodic Table.

Context: part (b) (introduction)

(b) Pelopium was the suggested name for a new element discovered in a mineral.

Pelopium was later found to be a mixture of niobium, Nb, and tantalum, Ta.

Only one naturally occurring isotope exists for each of Nb and Ta.

(i) Complete Table 1.1.

Table 1.1

isotope	relative isotopic mass	number of protons	number of neutrons
${}^{93}_{41}\text{Nb}$	92.91		
${}^{181}_{73}\text{Ta}$	180.95		

[2]

28. O/N 24/P22/Q3(c)

Context

3 Nitrogen and phosphorus are elements in Group 15 of the Periodic Table.

(c) Fig. 3.1 shows a sketch of some of the ionisation energies of phosphorus, P.

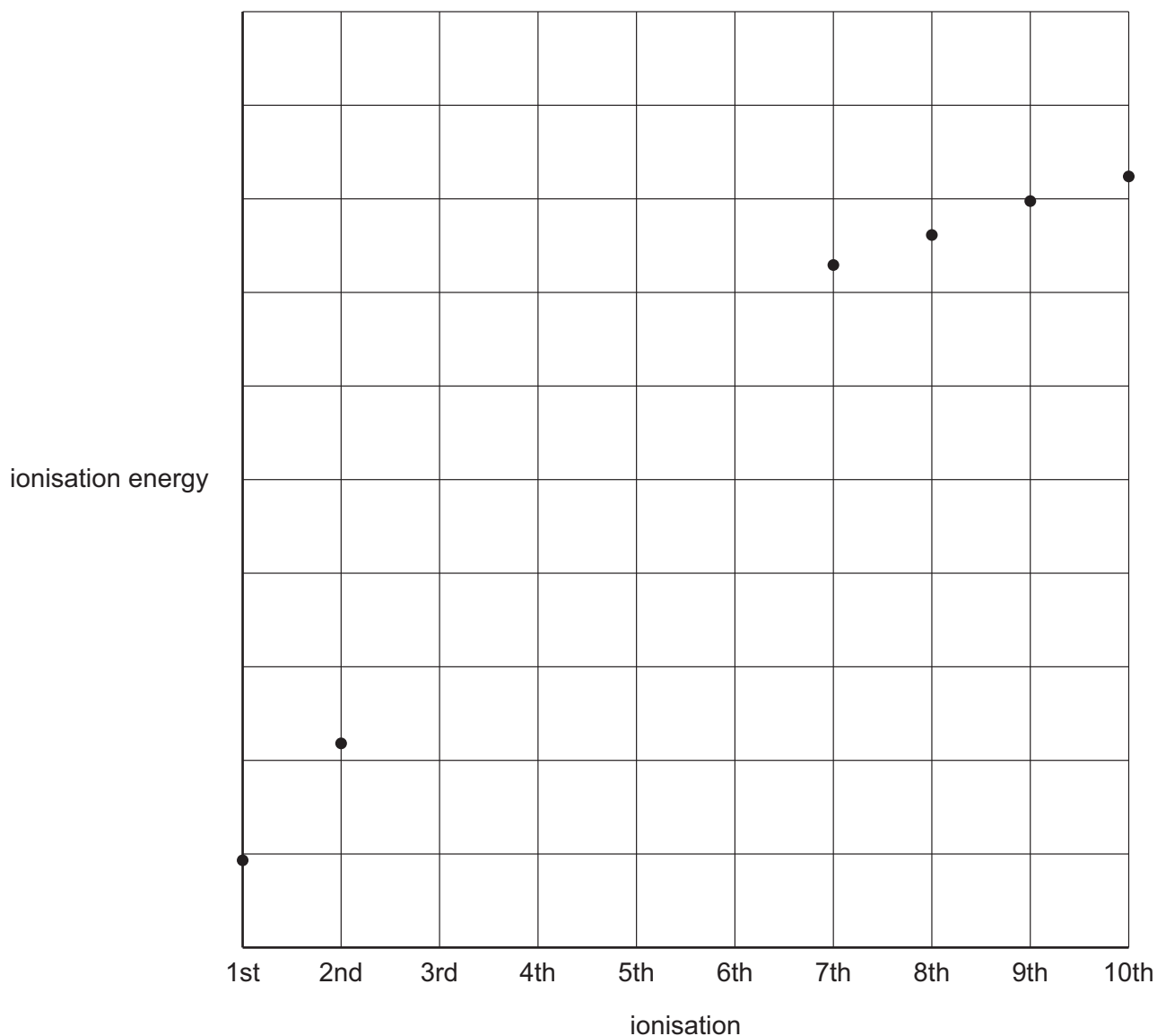


Fig. 3.1

(i) Construct an equation to represent the third ionisation energy of P.

[1]

(ii) Complete the graph in Fig. 3.1 to show the third to sixth ionisation energies of P.

[2]

29. O/N 24/P23/Q1(a)

Context

1 Cobalt, rhodium and iridium are metals in the same group of the Periodic Table.

(a) The shorthand electronic configuration of cobalt is $[\text{Ar}]3d^74s^2$.

(i) Identify what is meant by $[\text{Ar}]$ by giving its full electronic configuration.

[1]

(ii) The lowest-energy electrons in cobalt are in the 1s orbital.

Draw the shape of a 1s orbital.

[1]

(iii) Deduce the number of unpaired electrons in a cobalt atom.

[1]

30. O/N 24/P23/Q1(b)

Context

1 Cobalt, rhodium and iridium are metals in the same group of the Periodic Table.

(b) Table 1.1 gives some details of the stable naturally occurring isotopes of rhodium and iridium.

Table 1.1

isotope	number of protons	number of neutrons	total number of electron shells
$^{103}_{45}\text{Rh}$		58	
$^{191}_{77}\text{Ir}$			6
$^{193}_{77}\text{Ir}$			6

Complete Table 1.1.

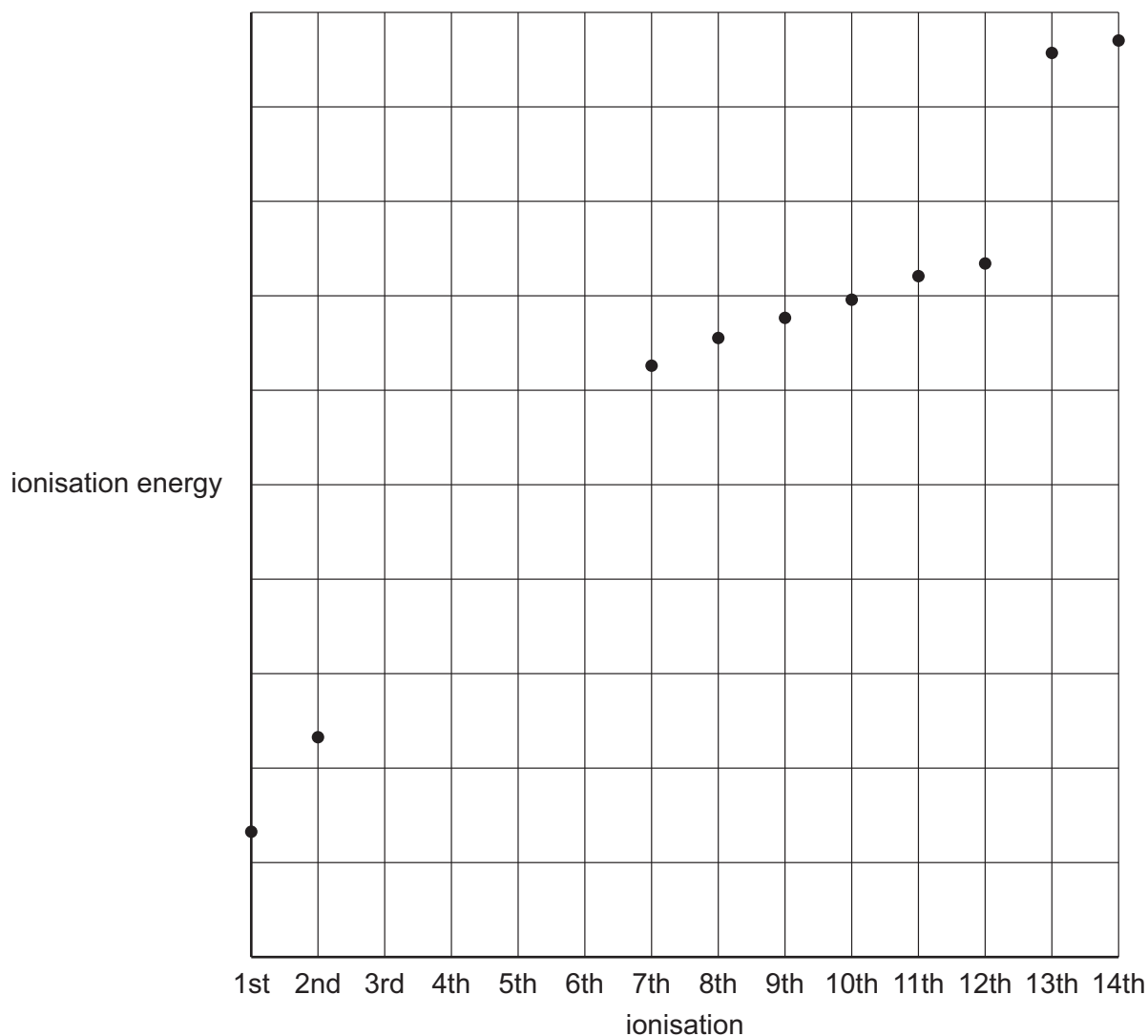
[3]

31. O/N 24/P23/Q3(c)

Context

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

(c) Fig. 3.1 shows a sketch of some of the ionisation energies of silicon, Si.

**Fig. 3.1**

(i) Complete the graph in Fig. 3.1 to show the third to sixth ionisation energies of Si.

[2]

(ii) Construct an equation to represent the second ionisation energy of Si.

[1]

32. M/J 24/P22/Q1(a)

1 (a) Complete Table 1.1 using relevant information from the Periodic Table.

Table 1.1

	nucleon number	proton number	number of electrons
Mg^{2+}	24		
Al^{3+}	27		

[2]

33. M/J 24/P22/Q1(b)

(b) State and explain the difference in the ionic radius of Al^{3+} compared to Mg^{2+} .

[3]

34. M/J 24/P22/Q3(c)

(c) Determine the full electronic configuration of Fe^{3+} .

[1]

35. O/N 23/P21/Q1(a)

Context

1 The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to Cl.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	Cl
number of electrons in 3p subshell			
total number of unpaired electrons			
first ionisation energy / kJ mol^{-1}	1060		1260
formula of most common anion	P^{3-}	S^{2-}	Cl^{-}

(a) (i) Complete Table 1.1 to show the number of electrons in the 3p subshell and the total number of unpaired electrons in an atom of P, S and Cl. [2]

(ii) Construct an equation to represent the first ionisation energy of P.

[1]

(iii) Three possible values for the first ionisation energy of S are given.

 1000 kJ mol^{-1} **1160 kJ mol^{-1}** **1320 kJ mol^{-1}**

Circle the correct value.

Explain your choice by comparing your chosen value to those of P and Cl.

[4]

36. O/N 23/P21/Q1(b)

Context

- 1** The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to Cl.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	Cl
number of electrons in 3p subshell			
total number of unpaired electrons			
first ionisation energy /kJ mol ⁻¹	1060		1260
formula of most common anion	P ³⁻	S ²⁻	Cl ⁻

- (b)** P³⁻, S²⁻ and Cl⁻ have the same number of electrons.

- (i)** Give the full electronic configuration of P³⁻.

[1]

- (ii)** State the trend in ionic radius shown by P³⁻, S²⁻ and Cl⁻.

Explain your answer.

[2]

37. O/N 23/P22/Q1(b)

also Unit 9 (1 mark), Unit 3 (1 mark); this unit 7 marks

Context

1 The elements silicon, phosphorus and sulfur are in Period 3 of the Periodic Table.

(b) Table 1.1 shows some properties of the elements Si to S.

The first ionisation energy of P is **not** shown.

Table 1.1

property	Si	P	S
total number of electrons in s subshells			
total number of electrons in p subshells			
first ionisation energy/kJ mol ⁻¹	786		1000
formula of most common chloride	SiCl ₄	PCl ₅	SCl ₂

(i) Complete Table 1.1 to show the total number of s and p electrons in an atom of Si, P and S.

[2]

(ii) Construct an equation to represent the first ionisation energy of Si.

[1]

(iii) Three possible values for the first ionisation energy of P are given.

619 kJ mol⁻¹

893 kJ mol⁻¹

1060 kJ mol⁻¹

Circle the correct value.

Explain your choice, including a comparison of your chosen value to those of Si and S.

[4]

(iv) SiCl₄ and PCl₅ each react with water, forming misty fumes.

Identify the chemical responsible for the misty fumes.

[1]

(v) Predict the shape of the SCl₂ molecule.

[1]

38. O/N 23/P23/Q1(a)

Context

- 1** The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to Cl.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	Cl
number of electrons in 3p subshell			
total number of unpaired electrons			
first ionisation energy /kJ mol ⁻¹	1060		1260
formula of most common anion	P ³⁻	S ²⁻	Cl ⁻

- (a) (i) Complete Table 1.1 to show the number of electrons in the 3p subshell and the total number of unpaired electrons in an atom of P, S and Cl. [2]
- (ii) Construct an equation to represent the first ionisation energy of P.

[1]

- (iii) Three possible values for the first ionisation energy of S are given.

1000 kJ mol⁻¹

1160 kJ mol⁻¹

1320 kJ mol⁻¹

Circle the correct value.

Explain your choice by comparing your chosen value to those of P and Cl.

[4]

39. O/N 23/P23/Q1(b)

Context

- 1** The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to Cl.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	Cl
number of electrons in 3p subshell			
total number of unpaired electrons			
first ionisation energy /kJ mol ⁻¹	1060		1260
formula of most common anion	P ³⁻	S ²⁻	Cl ⁻

- (b)** P³⁻, S²⁻ and Cl⁻ have the same number of electrons.

- (i)** Give the full electronic configuration of P³⁻.

[1]

- (ii)** State the trend in ionic radius shown by P³⁻, S²⁻ and Cl⁻.

Explain your answer.

[2]

40. M/J 23/P21/Q1(a)

Context

- 1** Tellurium is an element in Group 16. The most common isotope of tellurium is ¹³⁰Te. Its electronic configuration is [Kr] 4d¹⁰ 5s² 5p⁴.

- (a)** Complete Table 1.1.

Table 1.1

	nucleon number	number of neutrons	number of electrons
¹³⁰ Te			

[3]

41. M/J 23/P21/Q1(b)

Context

- 1** Tellurium is an element in Group 16. The most common isotope of tellurium is ¹³⁰Te. Its electronic configuration is [Kr] 4d¹⁰ 5s² 5p⁴.

- (b)** Identify the sub-shell in an atom of Te that contains electrons with the lowest energy.

[1]

42. M/J 23/P21/Q1(c)

Context

- 1 Tellurium is an element in Group 16. The most common isotope of tellurium is ^{130}Te . Its electronic configuration is $[\text{Kr}] 4d^{10} 5s^2 5p^4$.

(c) Construct an equation to represent the first ionisation energy of Te.

[1]

43. M/J 23/P21/Q1(d)

Context

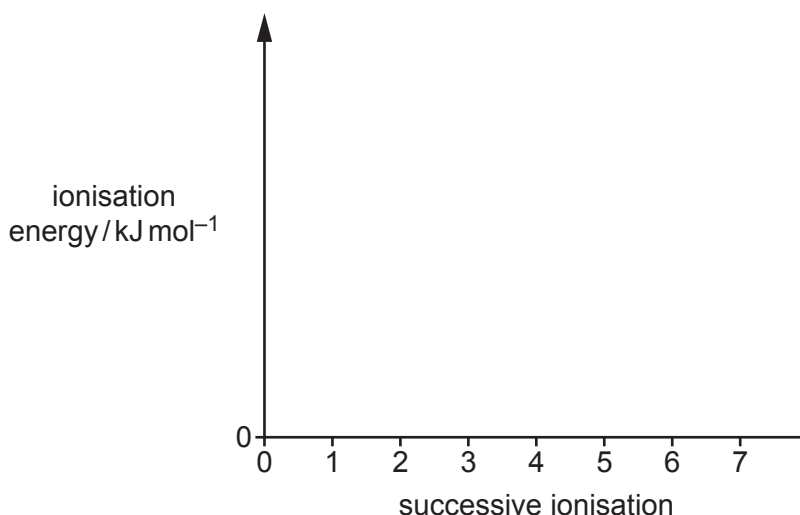
- 1 Tellurium is an element in Group 16. The most common isotope of tellurium is ^{130}Te . Its electronic configuration is $[\text{Kr}] 4d^{10} 5s^2 5p^4$.

(d) (i) The radius of Te ions decreases after each successive ionisation.

State **two** factors that are responsible for the increase in the first six ionisation energies of Te.

[2]

(ii) Sketch a graph in Fig. 1.1 to show the trend in the first **seven** ionisation energies of Te.

**Fig. 1.1**

[2]

44. M/J 23/P23/Q1(d)(iii)

Context

- 1 Copper is used in electrical equipment. It has a melting point of 1085°C .

Context: part (d) (introduction)

(d) When $\text{KI}(\text{aq})$ is added to $\text{CuSO}_4(\text{aq})$ the blue-coloured solution turns brown and a white precipitate of $\text{CuI}(\text{s})$ is seen.

The reaction between copper ions and iodide forms only two products.

(iii) State the full electronic configuration of Cu^{2+} .

[1]

45. MAR 23/P22/Q1(c)

Context

- 1** The Pauling electronegativity values of elements can be used to predict the chemical properties of compounds.

Use the information in Table 1.1 to answer the following questions.

Table 1.1

element	H	Li	C	O	S
Pauling electronegativity value	2.1	1.0	2.5	3.5	2.6
first ionisation energy/kJ mol ⁻¹	1310	519	1090	1310	1000
second ionisation energy/kJ mol ⁻¹	—	7300	2350	3390	2260

- (c) (i)** Write an equation that represents the first ionisation energy of H.

[1]

- (ii)** Explain why there is no information given in Table 1.1 for the second ionisation energy of H.

[1]

- (iii)** Give the full electronic configuration of S²⁺(g).

[1]

46. O/N 22/P21/Q1(a)(i)

Context

- 1** Atoms with nuclei containing an odd number of protons tend to have fewer isotopes than those with an even number of protons.

Context: part (a) (introduction)

- (a)** Gallium has two stable isotopes, ⁶⁹Ga and ⁷¹Ga.

- (i)** Complete Table 1.1 to show the numbers of protons, neutrons and electrons in the two stable isotopes of gallium.

Table 1.1

isotope	number of protons	number of neutrons	number of electrons
⁶⁹ Ga			
⁷¹ Ga			

[2]

47. O/N 22/P21/Q1(b)

Context

- 1** Atoms with nuclei containing an odd number of protons tend to have fewer isotopes than those with an even number of protons.

(b) Potassium also has two stable isotopes. Both isotopes have the same chemical properties.

- (i)** Explain why both isotopes of potassium have the same chemical properties.

[1]

- (ii)** State the full electronic configuration of an atom of potassium.

[1]

- (iii)** The first, second and third ionisation energies of potassium are 418, 3070 and 4600 kJ mol⁻¹, respectively.

Use this information to explain why potassium is in Group 1.

[2]

48. O/N 22/P22/Q1(b)

Context

- 1** Species such as NH₄⁺, CO₃²⁻ and PO₄³⁻ are examples of molecular ions.

- (b)** Complete Table 1.1 to show the total numbers of protons and electrons in the molecular ions NH₄⁺, CO₃²⁻ and PO₄³⁻.

Table 1.1

molecular ion	total number of protons	total number of electrons
NH ₄ ⁺		
CO ₃ ²⁻		
PO ₄ ³⁻		

[3]

49. O/N 22/P23/Q1(a)(i)

Context

- 1** Atoms with nuclei containing an odd number of protons tend to have fewer isotopes than those with an even number of protons.

Context: part (a) (introduction)

(a) Gallium has two stable isotopes, ^{69}Ga and ^{71}Ga .

- (i)** Complete Table 1.1 to show the numbers of protons, neutrons and electrons in the two stable isotopes of gallium.

Table 1.1

isotope	number of protons	number of neutrons	number of electrons
^{69}Ga			
^{71}Ga			

[2]

50. O/N 22/P23/Q1(b)

Context

- 1** Atoms with nuclei containing an odd number of protons tend to have fewer isotopes than those with an even number of protons.

(b) Potassium also has two stable isotopes. Both isotopes have the same chemical properties.

- (i)** Explain why both isotopes of potassium have the same chemical properties.

[1]

- (ii)** State the full electronic configuration of an atom of potassium.

[1]

- (iii)** The first, second and third ionisation energies of potassium are 418, 3070 and 4600 kJ mol⁻¹, respectively.

Use this information to explain why potassium is in Group 1.

[2]

51. M/J 22/P21/Q1(a)

Context

- 1** Calcium, magnesium and radium are Group 2 elements. Radium follows the same trends as the other members of Group 2.

- (a)** Identify the highest energy orbital which contains electrons in a calcium atom. Sketch the shape of this orbital.

identity of highest energy orbital in Ca

shape

[1]

52. M/J 22/P21/Q1(d)

Context

- 1** Calcium, magnesium and radium are Group 2 elements. Radium follows the same trends as the other members of Group 2.

- (d) (i)** $^{25}_{12}\text{Mg}$ is an isotope of magnesium.

Determine the number of protons and neutrons in an atom of $^{25}_{12}\text{Mg}$.

number of protons

number of neutrons

[1]

- (ii)** State the full electronic configuration of an atom of $^{25}_{12}\text{Mg}$.

[1]

53. M/J 22/P21/Q1(e)(iii)

Context

- 1** Calcium, magnesium and radium are Group 2 elements. Radium follows the same trends as the other members of Group 2.

Context: part (e) (introduction)

- (e)** A sample of magnesium contains three isotopes, ^{25}Mg , ^{26}Mg and **X**.

The percentage abundance of the three isotopes is shown in Table 1.1.

Table 1.1

isotope of Mg	mass/a.m.u.	percentage abundance/%
X		78.99
^{25}Mg	24.99	10.00
^{26}Mg	25.98	11.01

- (iii)** State **one** similarity and **one** difference in the properties of these isotopes of magnesium. Explain your answer.

[2]

54. M/J 22/P22/Q4(b)(iii)

Context: part (b) (introduction)

(b) Halothane is an anaesthetic.

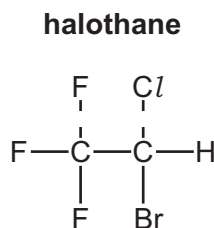


Fig. 4.1

(iii) Complete Fig. 4.2 to show the arrangement of electrons in a bromine atom using the electrons in boxes notation.

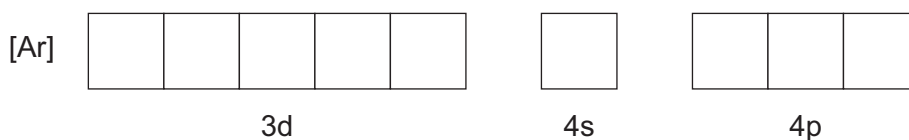


Fig. 4.2

[1]

55. M/J 22/P23/Q1(a)

1 (a) Define first ionisation energy.

[2]

56. M/J 22/P23/Q1(b)

(b) Successive ionisation energies for element **A** are shown in Table 1.1.

Table 1.1

ionisation	1st	2nd	3rd	4th	5th	6th	7th	8th
ionisation energy/kJ mol ⁻¹	1310	3390	5320	7450	11 000	13 300	71 000	84 100

Use Table 1.1 to deduce the group of the Periodic Table that **A** belongs to. Explain your answer.

Group

[1]

57. M/J 22/P23/Q1(c)

(c) Across Period 3 there is a general trend for first ionisation energies to increase due to the increase in attraction between the nucleus and the outer electron.

Explain why the first ionisation energy of sulfur is less than the first ionisation energy of phosphorus.

[2]

58. M/J 22/P23/Q1(d)

(d) In an Al^{2+} ion the nuclear attraction for the outer electron is stronger than in an atom of Na.

Compare the electronic structures of Al^{2+} and an atom of Na and explain why the third ionisation energy of aluminium is greater than the first ionisation energy of sodium.

[2]

59. M/J 22/P23/Q1(e)

(e) An isotope of copper has a relative isotopic mass of 65.

Complete Table 1.2 for an atom of copper-65.

Table 1.2

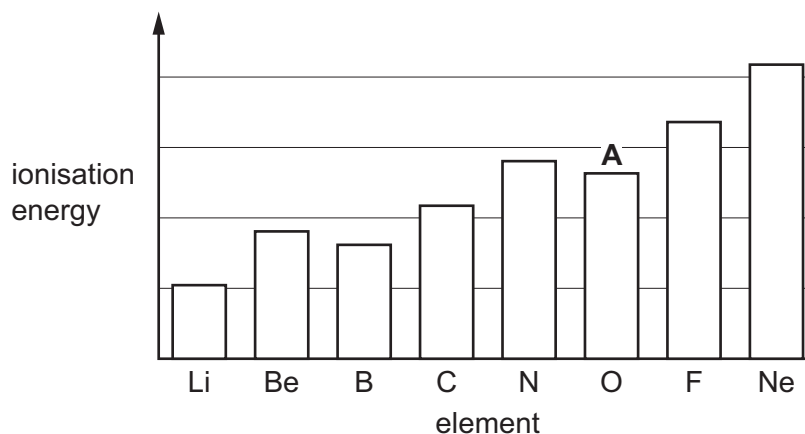
	atomic number	nucleon number	number of neutrons	electronic arrangement
copper-65				

[3]

60. MAR 22/P22/Q1(a)

Context

1 Fig. 1.1 shows how **first** ionisation energies vary across Period 2.

**Fig. 1.1**

(a) Construct an equation to represent the **first** ionisation energy of oxygen. Include state symbols.

[1]

61. MAR 22/P22/Q1(b)

Context

- 1** Fig. 1.1 shows how **first** ionisation energies vary across Period 2.

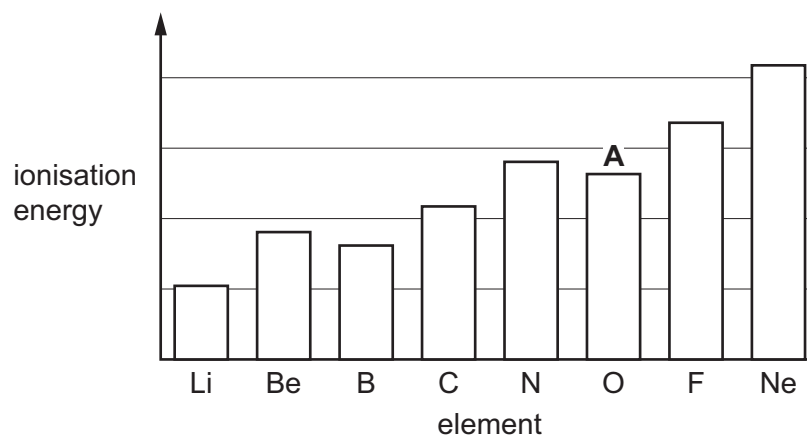


Fig. 1.1

- (b) (i)** State and explain the general trend in first ionisation energies across Period 2.

[3]

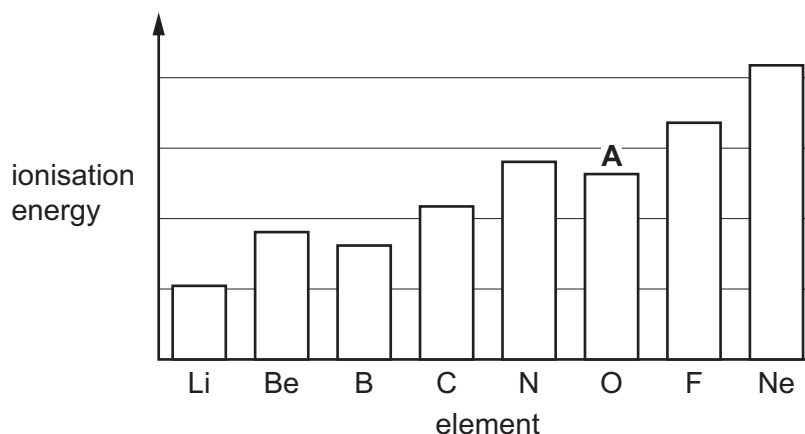
- (ii)** Explain why ionisation energy **A** in Fig. 1.1 does **not** follow the general trend in first ionisation energies across Period 2.

[2]

62. MAR 22/P22/Q1(c)

Context

- 1** Fig. 1.1 shows how **first** ionisation energies vary across Period 2.

**Fig. 1.1**

- (c) Element **E** is in Period 3 of the Periodic Table.
The first eight ionisation energy values of **E** are shown in Table 1.1.

Table 1.1

ionisation	1st	2nd	3rd	4th	5th	6th	7th	8th
ionisation energy/kJ mol ⁻¹	577	1820	2740	11 600	14 800	18 400	23 400	27 500

Deduce the full electronic configuration of **E**.
Explain your answer.

full electronic configuration of **E** =

explanation

[3]

Answers Section

1. M/J 26/P21/Q1(b)

1(b)(i)	 AND label 1s	1	I n in $1s^n$ I spherical I axes I dot-and-cross diagrams
1(b)(ii)	$\text{Cl(g)} \rightarrow \text{Cl}^+(g) + e^{(-)}$	1	A $\text{Cl(g)} - e^{(-)} \rightarrow \text{Cl}^+(g)$ I $\text{Cl(g)} + e^{(-)} \rightarrow \text{Cl}^+(g) + 2 e^{(-)}$
1(b)(iii)	M1 (small) increase in shielding in Cl is outweighed by M2 increased nuclear charge in Cl M3 Cl has greater nuclear attraction OR M1 similar shielding M2 Cl has greater nuclear charge M3 Cl has greater nuclear attraction	ora ora	3 For all 3 marks to be awarded: 'chlorine' needs only to be identified once in M1 or M2 or M3 chlorine can be identified by name / symbol or trend across a period e.g. number of protons increases across the period M1 great(er) = increase = high(er) = more = large(r) etc for shielding: similar = constant = equal = same shielding = screening I electron removed from same (sub) shell I same number of shells I same electron cloud I references to arrangement of electrons in the orbitals I atomic radius references M2 A number of protons for nuclear charge A greater atomic number for greater nuclear charge A Cl has one more proton than S I 'effective' of effective nuclear charge I nucleon number M3 (nuclear) attraction = pull = draws (into nucleus) For M2 and M3 'nuclear' needs only to be mentioned once e.g. greater nuclear attraction may be implied by reference to 'increased attraction due to increased nuclear charge' owtte If M2 and M3 not awarded: Max 1 for greater nuclear attraction linked to greater nuclear charge if species not identified or incorrectly identified

2. M/J 26/P22/Q1(a)

1(a)(i)	4 / four	1	BOD ($1s^2 2s^2$) $2p^4$
1(a)(ii)	2p AND 	1	A $2p^4$ for 2p I p alone I shading but it must have a node I axes I orientation I ignore subscripts $p_x p_y p_z$

3. M/J 26/P22/Q2(a)

2(a)(i)	curve x = electron AND curve y = proton	1	
---------	--	----------	--

2(a)(ii)	particles in y are deflected less because they have larger mass	1	ecf name given for particles in curve y have the heavier mass. Assume 'they' = y = particles of y unless explicitly refer to 'x' in answer A they are heavier A higher mass / charge ratio A greater weight A they have greater momentum A (particles in) x are lighter I 'electrons have negligible mass' (comparison required) I lower charge density I density I protons are larger I kinetic energy R protons have greater charge density
----------	---	---	---

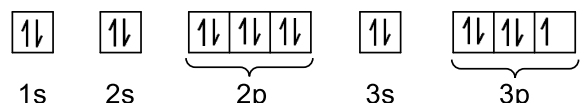
4. M/J 26/P22/Q2(b)

2(b)(i)	(atom) A and (atom) B OR (atom) A and (atom) C	1	A 6 and 17 OR 6 and 14 A carbon and chlorine OR carbon and silicon
2(b)(ii)	(atom) B and (atom) C	1	A 17 and 14 A chlorine and silicon
2(b)(iii)	M1 atom A has (one) fewer (electron) shell(s) (so decreased shielding effect) M2 (so) more attraction of nucleus for outer/valence electrons / outer or valent shell OR In atom C the effect of more protons / increased nuclear charge is less significant than / outweighed by the (increased shielding) effect of the additional shell ora = ✓✓	2	Subject of question is A, so 'it' = A A reverse argument for atom C I subshells / orbitals I incorrect identities of atoms A, B and C M1 A more shells down group shell = energy level = quantum level If number of shells is given, it must be correct (atom A=2 and atom C=3) M1 A atom A has outer electrons in a lower energy shell M1 I distance from the nucleus M1 I reference to one fewer outer/valence shell M2 A more attraction of protons to the outer/valence electrons M2 BOD pull for attraction

5. M/J 26/P23/Q2(a)

2(a)	(down the group) M1 (atomic radius) increases OR (atom) volume increases OR Ne < Ar < Kr < Xe M2 (down the group) the number of shells (of electrons) increases	2	shells = energy levels I number of sub-shells increases I nuclear attraction of outer electron decreases I increased shielding effect outweighs the increase in nuclear charge I outer electron further way
------	---	---	--

6. M/J 26/P23/Q5(b)(i)

5(b)(i)	 1s 2s 2p 3s 3p	1	A any arrangement of 2 pairs and one single electron in 3 p A double headed arrows.
---------	---	---	---

7. M/J 26/P24/Q1(a)

1(a)(i)	atom or ion	number of protons	number of neutrons	2	●✓✓✓
	^{27}Al	13	14		
	$^{28}\text{Al}^{3+}$	13	15		

1(a)(ii)		1	A single or double headed arrows A arrows in any direction R lines instead of arrows
----------	--	---	---

8. MAR 26/P22/Q1(a)

1(a)(i)	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$	1
1(a)(ii)	(ionisation energy decreases because of) - more shells / larger (atomic/ionic) radius - increased shielding - less attraction (for the nucleus/protons) / lower energy required (to remove electrons)	2
1(a)(iii)	$\text{Ba}^{2+}(\text{g}) \rightarrow \text{Ba}^{3+}(\text{g}) + \text{e}^{-}$	1
1(a)(iv)	M1 electron removed from closer / inner / lower-energy (p sub)shell/orbital M2 (much) less shielding OR greater nuclear attraction	2

9. M/J 25/P21/Q2(a)

2(a)	atoms (of an element) with same number of protons and different number of neutrons	1
------	--	---

10. M/J 25/P21/Q2(b)(ii)

2(b)(ii)	particle	number of particles in one atom of ^{56}Fe	2
	protons	26	
	nucleons	56	

11. M/J 25/P21/Q2(c)

2(c)	5	1
------	---	---

12. M/J 25/P21/Q2(d)

2(d)	$\text{Fe}(\text{g}) \rightarrow \text{Fe}^{+}(\text{g}) + \text{e}^{-}$	1
------	--	---

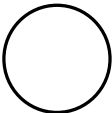
13. M/J 25/P21/Q2(e)

2(e)	M1 same electronic configuration OR same shielding M2 same number of protons OR same nuclear charge M3 so same nuclear attraction to outer shell electron M4 no change / will not be affected	4
------	--	---

14. M/J 25/P22/Q2(b)

2(b)	1 (pair)	1
------	----------	---

15. M/J 25/P22/Q2(c)

2(c)		1
------	---	---

16. M/J 25/P22/Q2(d)

2(d)	species present in an ion of 56-Fe^{3+}	number	2
	electrons	M1 23	
	M2 neutrons	30	

17. M/J 25/P22/Q2(e)

2(e)	M1 smaller / decreases M2 loss of outer / valence shell (of electrons) OR loss of electrons so increase in (force of) attraction from nucleus OR loss of electrons so increase nuclear attraction	2
------	--	---

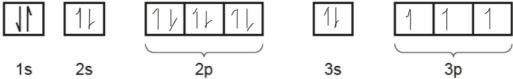
18. M/J 25/P23/Q1(a)

1(a)(i)	$1s^2 2s^2 2p^5$	1
1(a)(ii)	2 (pairs)	1
1(a)(iii)		1

19. M/J 25/P23/Q1(b)

1(b)(i)	$S(g) \rightarrow S^+(g) + e^{(-)}$ OR $S(g) - e^{(-)} \rightarrow S^+(g)$	1
1(b)(ii)	M1 pair of electrons in a 3p orbital (in S) repel each other M2 outweighs the effect of increase in nuclear charge	2
1(b)(iii)	M1 $Na^+ < Ne < F^-$ M2 (all have) same number of electrons / isoelectronic M3 larger species has smaller nuclear charge / fewer protons M4 (so) smaller nuclear attraction (to outer electrons)	4

20. M/J 25/P24/Q1(c)

1(c)(i)	$Ar^{2+}(g) \rightarrow Ar^{3+}(g) + e^{(-)}$ M1 a correct equation which represents an ionisation process e.g. $X^{n+} \rightarrow X^{(n+1)+} + e^{(-)}$ M2 correct equation to show third ionisation energy of Ar including (g) state symbols	2
1(c)(ii)	M1 nuclear charge M2 distance of outer electron(s) from nucleus OR ionic radius OR shielding (by inner shells and / or sub-shells)	2
1(c)(iii)	M1 (most significant factor) = distance of outer electron(s) from nucleus OR ionic radius OR shielding OR shells M2 (3rd IE) value for (Na and) Mg is (much) greater than Al (to Ar) OR (Na and) Mg has electron(s) removed from a shell that is closer to the nucleus / less shielding OR (Na and) Mg has electron(s) removed from a shell that is lower in energy	2
1(c)(iv)		1

21. MAR 25/P22/Q2(b)

2(b)(i)	ionic radius increases / ions larger (down the group) extra shell (of electrons)	2
2(b)(ii)	ionisation energies increase because less shielding / successive ionic radii are smaller / greater attraction (of electron) to nucleus (large difference between 2nd and 3rd IE) because 3rd e ⁻ coming from an inner shell / lower energy level	2

22. MAR 25/P22/Q2(c)

2(c)(i)	most of the mass is in the nucleus because protons and neutrons / nucleons are found in the nucleus	2												
2(c)(ii)	<table border="1"> <thead> <tr> <th>isotope</th><th>number of protons</th><th>number of neutrons</th></tr> </thead> <tbody> <tr> <td>magnesium-24</td><td>12</td><td>12</td></tr> <tr> <td>magnesium-25</td><td>12</td><td>13</td></tr> <tr> <td>magnesium-26</td><td>12</td><td>14</td></tr> </tbody> </table>	isotope	number of protons	number of neutrons	magnesium-24	12	12	magnesium-25	12	13	magnesium-26	12	14	2
isotope	number of protons	number of neutrons												
magnesium-24	12	12												
magnesium-25	12	13												
magnesium-26	12	14												
2(c)(iii)	neutron beam horizontal straight line from same source electron beam downwards curving line from same source line deflected more than proton beam	3												

23. O/N 24/P21/Q1(a)

1(a)(i)	1s ² 2s ² 2p ⁶ 3s ² 3p ⁶	1
1(a)(ii)		1
1(a)(iii)	3 (three)	1

24. O/N 24/P21/Q1(b)

1(b)					3
	isotope	number of protons	number of neutrons	total number of electron shells	
	$^{103}_{45}\text{Rh}$	● 45	58	● 5	
	$^{191}_{77}\text{Ir}$	● 77	● 114	6	
	$^{193}_{77}\text{Ir}$	● 77	● 116	6	

25. O/N 24/P21/Q3(c)

3(c)(i)	M1 general increase	1
	M2 increase in IE between IE4 and IE5 is noticeably the largest increase between IE's shown	1
3(c)(ii)	Si ⁺ (g) → Si ²⁺ (g) + e ⁻	1

26. O/N 24/P22/Q1(a)

1(a)(i)	lowest energy state owtte examples • (with) lowest energy • (with) no external energy • (with) no added energy • (where) electrons are not promoted to a higher energy level • most energetically stable	1						
1(a)(ii)	[Ar] <table><tr><td>↑</td><td>↑</td><td>↑</td><td></td><td></td><td>↑↓</td></tr></table> <i>The 3 × d electrons may go in any of the d orbitals</i>	↑	↑	↑			↑↓	1
↑	↑	↑			↑↓			
1(a)(iii)	12	1						

27. O/N 24/P22/Q1(b)(i)

1(b)(i)	isotope	RIM	# protons	# neutrons	2
	$^{93}_{41}\text{Nb}$	92.91	● 41	● 52	
	$^{181}_{73}\text{Ta}$	180.95	● 73	● 108	

28. O/N 24/P22/Q3(c)

3(c)(i)	$\text{P}^{2+}(\text{g}) \rightarrow \text{P}^{3+}(\text{g}) + \text{e}^{-}$	1
3(c)(ii)	general increase	1
	increase in IE between IE5 and IE6 is noticeably the largest increase between IEs shown	1

29. O/N 24/P23/Q1(a)

1(a)(i)	$1s^2 2s^2 2p^6 3s^2 3p^6$	1
1(a)(ii)		1
1(a)(iii)	3 (three)	1

30. O/N 24/P23/Q1(b)

1(b)					3
	isotope	number of protons	number of neutrons	total number of electron shells	
	$^{103}_{45}\text{Rh}$	● 45	58	● 5	
	$^{191}_{77}\text{Ir}$	● 77	● 114	6	
	$^{193}_{77}\text{Ir}$	● 77	● 116	6	

31. O/N 24/P23/Q3(c)

3(c)(i)	M1 general increase	1
	M2 increase in IE between IE4 and IE5 is noticeably the largest increase between IE's shown	1
3(c)(ii)	$\text{Si}^+(\text{g}) \rightarrow \text{Si}^{2+}(\text{g}) + \text{e}^{-}$	1

32. M/J 24/P22/Q1(a)

1(a)					2
	Mg ²⁺	24	12	10	
	Al ³⁺	27	13	10	
4 correct = 2 marks 2 or 3 correct = 1 mark 0 or 1 correct = 0 marks					

33. M/J 24/P22/Q1(b)

1(b)	M1 it / Al ³⁺ is smaller (compared to Mg ²⁺) M2 greater nuclear attraction for remaining electrons M3 same shielding effect AND greater nuclear charge	3
------	--	----------

34. M/J 24/P22/Q3(c)

3(c)	full electronic configuration of Fe ³⁺ 1s ² 2s ² 2p ⁶ 3s ² 3p ⁶ 3d ⁵	1
------	---	----------

35. O/N 23/P21/Q1(a)

1(a)(i)	total # 3p e ⁻	3	4	5	1
	total # unpaired e ⁻	3	2	1	1
1(a)(ii)	P(g) → P ⁺ (g) + e ⁽⁻⁾				1
1(a)(iii)	1000 (kJmol ⁻¹)				1
	(S) less / least / lowest nuclear attraction (to outer electrons) (than P OR Cl) OR nuclear attraction for Cl is stronger (compared to S (and or P))				1
	<u>S less nuclear charge than Cl</u> OR <u>Cl has a greater nuclear charge than S</u>				1
	(S has a greater nuclear charge (of outer electrons) than P BUT) S has two electrons in a (3) p orbital AND resulting in spin-pair repulsion				1

36. O/N 23/P21/Q1(b)

1(b)(i)	1s ² 2s ² 2p ⁶ 3s ² 3p ⁶				1
1(b)(ii)	- decreases (from P³⁻ to Cl⁻) - increased attraction (of outer e⁻) for nucleus owtte - increased nuclear charge - same shielding •√•√				2

37. O/N 23/P22/Q1(b)

1(b)(i)	total e ⁻ in s subshell	6	6	6	1
	total e ⁻ in p subshell	8	9	10	1
1(b)(ii)	Si(g) → Si ⁺ (g) + e ⁻				1

1(b)(iii)	1060 (is circled)	1
	P / it has greater attraction of nucleus for outer electrons compared to Si ORA	1
	P / it has greater nuclear charge than Si ORA	1
	S has two electrons in a (3)p orbital AND resulting in spin-pair repulsion	1
1(b)(i)(v)	hydrogen chloride / HCl	1
1(b)(v)	non-linear	1

38. O/N 23/P23/Q1(a)

1(a)(i)	total # 3p e^-	3	4	5	1
	total # unpaired e^-	3	2	1	1
1(a)(ii)	$\text{P(g)} \rightarrow \text{P}^+(\text{g}) + e^{(-)}$				1
1(a)(iii)	1000 (kJ mol^{-1})				1
	(S) less / least / lowest nuclear attraction (to outer electrons) (than P OR Cl) OR nuclear attraction for Cl is stronger (compared to S (and or P))				1
	<u>S less nuclear charge than Cl</u> OR <u>Cl has a greater nuclear charge than S</u>				1
	(S has a greater nuclear charge (of outer electrons) than P BUT) S has two electrons in a (3)p orbital AND resulting in spin-pair repulsion				1

39. O/N 23/P23/Q1(b)

1(b)(i)	$1s^2 2s^2 2p^6 3s^2 3p^6$	1
1(b)(ii)	- decreases (from P^{3-} to Cl^-) - increased attraction (of outer e^-) for nucleus owtte - increased nuclear charge - same shielding $\bullet \sqrt{\bullet} \sqrt{\bullet}$	2

40. M/J 23/P21/Q1(a)

1(a)			nucleon number	number of neutrons	number of electrons	3
		Tellurium-130	130	78	52	

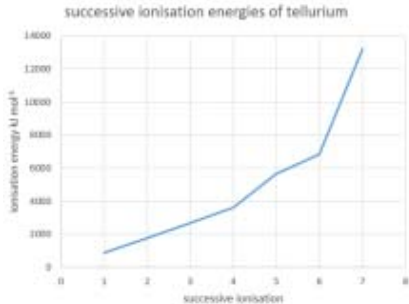
41. M/J 23/P21/Q1(b)

1(b)	1 s	1
------	-----	----------

42. M/J 23/P21/Q1(c)

1(c)	$\text{Te(g)} \rightarrow \text{Te}^+(\text{g}) + e^{(-)}$	1
------	--	----------

43. M/J 23/P21/Q1(d)

1(d)(i)	any two factors responsible for increase in IE in first 6 IE from: (increased) nuclear attraction / attraction for nucleus (decreased) shielding (by sub-shells) (IE 4 to 5) - spin-pair repulsion (IE 1 to 2)	2
1(d)(ii)	 M1 general increase M2 jumps from 4 to 5 and (largest) 6 to 7	2

44. M/J 23/P23/Q1(d)(iii)

1(d)(iii)	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^9 (4s^0)$	1
-----------	--	----------

45. MAR 23/P22/Q1(c)

1(c)(i)	$H(g) \rightarrow H^+(g) + e^-$	1
1(c)(ii)	H (cannot undergo second ionisation because it only) has one electron / H^+ has no electron	1
1(c)(iii)	$1s^2 2s^2 2p^6 3s^2 3p^2$	1

46. O/N 22/P21/Q1(a)(i)

1(a)(i)	columns 1 & 3 identical				1
	isotope	No of p's	No of n's	No of e's	
	⁶⁹ Ga	31	38	31	
	⁷¹ Ga	31	40	31	
	● √ √				1

47. O/N 22/P21/Q1(b)

1(b)(i)	they have the same electron arrangement / electronic configuration	1
1(b)(ii)	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$	1
1(b)(iii)	M1 big increase in IE between first and second M2 second (and third) electron(s) is removed from inner shell OR second (and third) electron(s) is removed from a shell closer to the nucleus OR second (and third) electron(s) has a stronger nuclear attraction ora	2

48. O/N 22/P22/Q1(b)

1(b)	NH_4^+ 11 p ⁺ AND 10 e ⁻	1
	CO_3^{2-} 30 p ⁺ AND 32 e ⁻	1
	PO_4^{3-} 47 p ⁺ AND 50 e ⁻	1

49. O/N 22/P23/Q1(a)(i)

1(a)(i)	columns 1 & 3 identical				1
	isotope	No of p's	No of n's	No of e's	
	⁶⁹ Ga	31	38	31	
	⁷¹ Ga	31	40	31	
	● √ √				1

50. O/N 22/P23/Q1(b)

1(b)(i)	they have the same electron arrangement / electronic configuration	1
1(b)(ii)	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$	1
1(b)(iii)	M1 big increase in IE between first and second M2 second (and third) electron(s) is removed from inner shell OR second (and third) electron(s) is removed from a shell closer to the nucleus OR second (and third) electron(s) has a stronger nuclear attraction ora	2

51. M/J 22/P21/Q1(a)

1(a)	Identify and draw the shape of highest energy orbital of Ca 4s AND 	1
------	--	----------

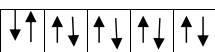
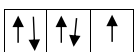
52. M/J 22/P21/Q1(d)

1(d)(i)	number of protons: 12 number of neutrons: 13	1
1(d)(ii)	$1s^2 2s^2 2p^6 3s^2$	1

53. M/J 22/P21/Q1(e)(iii)

1(e)(iii)	M1 (magnesium isotopes have) identical chemical properties AND same electron(ic) arrangement / configuration M2 different physical properties AND different number of neutrons	2
-----------	--	----------

54. M/J 22/P22/Q4(b)(iii)

4(b)(iii)	[Ar]   	1
-----------	--	----------

55. M/J 22/P23/Q1(a)

1(a)	- energy required - when one electron is removed - from each atom in one mole of - gaseous atoms two or three points for one mark, four points for two marks	2
------	---	----------

56. M/J 22/P23/Q1(b)

1(b)	Group VI / 16 AND large increase (in IE) after 6th	1
------	---	----------

57. M/J 22/P23/Q1(c)

1(c)	M1 reference to spin pair repulsion in (3)p orbital (in S) OR due to repulsion of two electrons in a (3)p orbital (in S) M2 outweighs increased nuclear charge (in S)	2
------	--	----------

58. M/J 22/P23/Q1(d)

1(d)	M1 <i>similarity in electronic structure / shielding of Al^{2+} and Na</i> both remove electron from (3)s ¹ / single electron in (3)s (sub-level / orbital) OR Al^{2+} and Na have same electronic configuration OR shielding (of outer electron) is the same M2 <i>greater nuclear charge / number of protons</i> Al^{2+} has greater nuclear charge OR 13p compared to 11p	2
------	---	----------

59. M/J 22/P23/Q1(e)

1(e)		atomic no.	nucleon no.	no.of neutrons	electronic arrangement	3
	copper –65	29	65	65 – 29 = 36	1s ² 2s ² 2p ⁶ 3s ² 3p ⁶ 3d ¹⁰ 4s ¹	
	M1 29 AND 65 M2 nucleon no – atomic no ALLOW ecf from M1 M3 electronic arrangement					

60. MAR 22/P22/Q1(a)

1(a)	$O(g) \rightarrow O^+(g) + e^-$	1
------	---------------------------------	----------

61. MAR 22/P22/Q1(b)

1(b)(i)	increase across period AND increased nuclear attraction for (valence / outer) electrons [1] increase in (positive) nuclear charge / number of protons (in the nucleus) [1] similar shielding (of outer electrons) [1]	3
1(b)(ii)	spin-pair repulsion (of electrons) in (2)p <u>orbital</u> [1] outweighs increased nuclear charge [1]	2

62. MAR 22/P22/Q1(c)

1(c)	$1s^2 2s^2 2p^6 3s^2 3p^1$ [1] greatest jump between 3rd and 4th ionisations [1] indicates three electrons in outer shell [1]	3
------	---	----------